

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	SWTID-2026-2085
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throught the code	Yes	Proper indentation maintained in all Python, HTML, CSS, and JavaScript files.
2	Proper File Structure	Files and folders are logically organize	Yes	Proper indentation maintained in all Python, HTML, CSS, and JavaScript files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names such as temperature, humidity, crop_prediction, and soil_data are used.
4	Function / Method Names	Functions are descriptively named	Yes	Descriptive variable names such as temperature, humidity, crop_prediction, and soil_data are used.
5	Code Comments	Inline and block comments explain log	Yes	Descriptive variable names such as temperature, humidity, crop_prediction, and soil_data are used.
6	Modular Design	Code is split into reusable functions/m	Yes	Separate modules for data preprocessing, model training, prediction, and Flask application.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused variables and duplicate code removed to improve maintainability.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling implemented for invalid or missing user inputs.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Flask Web Application	Python, Flask	Handles routing and user requests	High

2	Machine Learning Prediction Module	Python, Scikit-learn	Generates crop recommendations	High
3	Data Preprocessing Module	Python, Pandas, NumPy	Cleans and prepares input data Reused across multiple web pages	High
4	HTML Templates	HTML, CSS, Bootstrap	Reused across multiple web pages	Medium
5	Trained Model (.pkl)	Scikit-learn, Pickle	Used for repeated crop predictions	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder structure with clear separation of frontend, backend, model, and dataset.
Readability	5	Clean, readable code with meaningful naming conventions and formatting.
Reusability	5	Modular functions and reusable components improve maintainability and scalability.
Documentation / Comments	4	Sufficient comments provided to explain key sections and workflows.
Overall Score	4.8 / 5	The project follows good coding practices and is easy to understand, maintain, and extend.